



Frontotemporal Disorders

Information for Patients,
Families, and Caregivers

LEARN ABOUT:

- Frontotemporal dementia
- Primary progressive aphasia
- Movement disorders



National Institute on Aging
**National Institute of Neurological
Disorders and Stroke**

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Introduction to Frontotemporal Dementia

The symptoms of frontotemporal dementia (FTD) and other frontotemporal disorders gradually rob people of basic abilities — thinking, talking, walking, and socializing — that most of us take for granted. FTD and other frontotemporal disorders are a common cause of early-onset dementia. They often strike people in the prime of life when they are working and raising families. Families suffer, too, as they struggle to cope with the person's daily needs as well as changes in relationships and responsibilities.

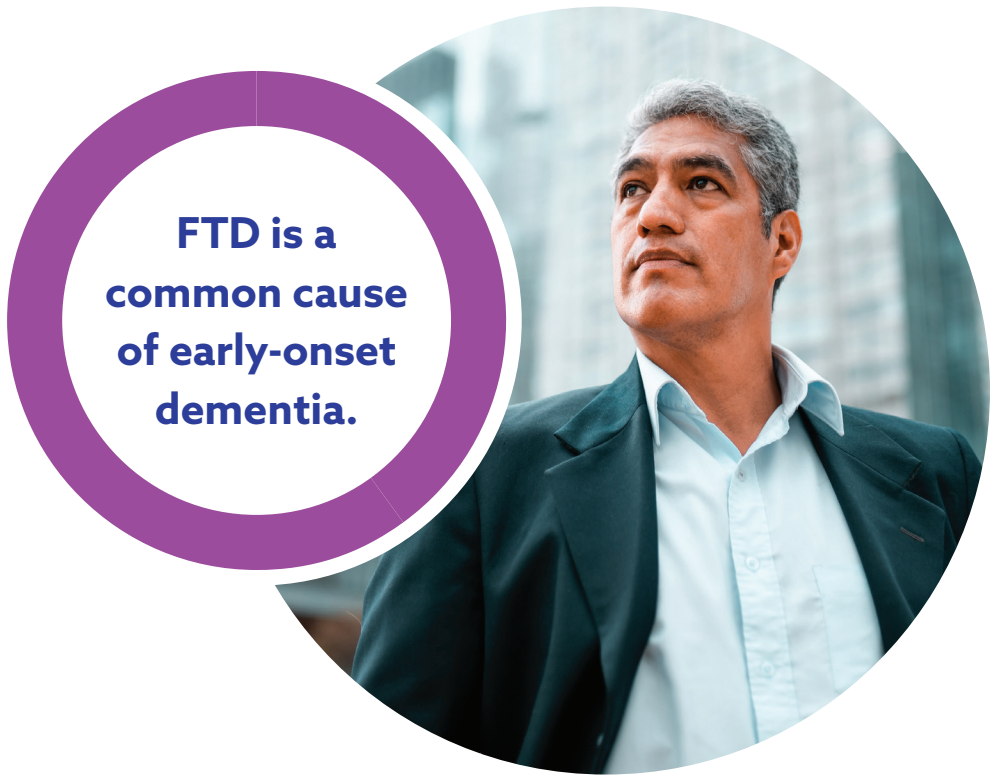
There is currently no cure for FTD or related disorders, and no treatments to slow or stop the progression of the disease, but there are ways to help manage the symptoms. Research is improving our awareness and understanding of these challenging conditions. This progress is opening doors to better diagnosis, improved care, and eventually, new treatments.

This booklet is meant to help people with frontotemporal disorders, their families, and caregivers learn more about these conditions and resources for coping. It explains what is known about the different types of disorders and how they are diagnosed. It also describes how frontotemporal disorders are treated and managed, with practical advice for caregivers.

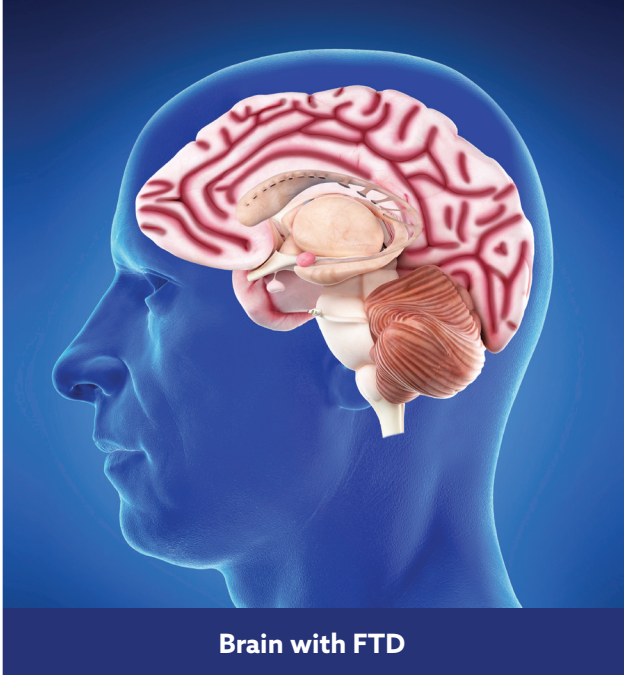
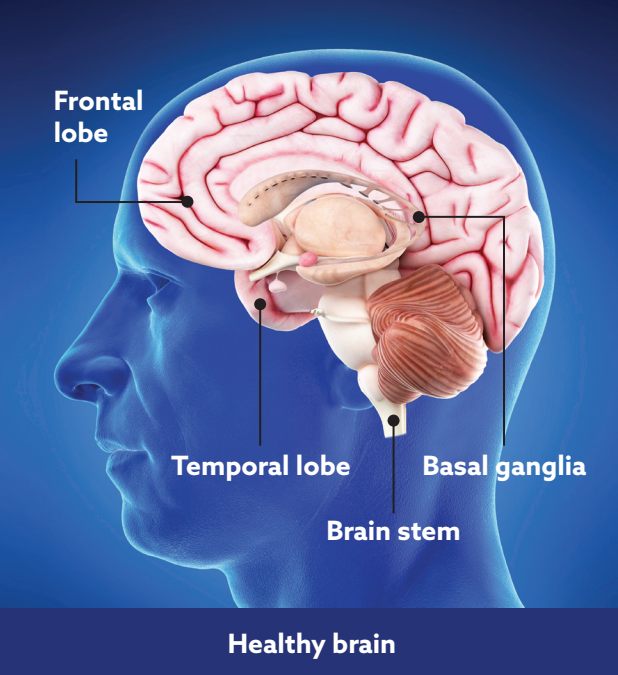
Frontotemporal disorders are the result of damage to neurons primarily in the frontal and temporal lobes of the brain. As neurons die in these regions, the frontal and temporal lobes atrophy, or shrink. Gradually, this damage causes difficulties in thinking and

behaviors normally controlled by these parts of the brain. Many possible symptoms can result, including unusual behaviors, emotional problems, trouble communicating, difficulty with work, or difficulty with movement and walking. FTD and related disorders are rare and tend to occur at a younger age than other forms of dementia. Roughly 60% of people with FTD are 45 to 64 years old.

FTD is progressive, meaning symptoms get worse over time. In the early stages, people may have just one type of symptom. As the disease progresses, other symptoms will appear as more parts of the brain are affected. It is difficult to predict how long someone with FTD will live. Some people live more than 10 years after diagnosis, while others live less than two years after they are diagnosed.



**FTD is a
common cause
of early-onset
dementia.**



FTD-Related Changes in the Brain

Frontotemporal disorders affect the frontal and temporal lobes of the brain. They can begin in the frontal lobe, the temporal lobe, or both. Initially, frontotemporal disorders leave other brain regions untouched, including those that control short-term memory.

The **frontal lobes**, situated above the eyes and behind the forehead, direct executive functioning. This includes planning and sequencing (thinking through which steps come first, second, third, and so on), prioritizing (doing more important activities first and less important activities last), multitasking (shifting from one activity to another as needed), and monitoring and correcting errors.

When functioning well, the frontal lobes also help manage emotional responses. They help people avoid inappropriate social behaviors, such as shouting loudly in a library or at a funeral. They aid in making decisions that make sense for a given situation. When the frontal lobes are damaged, people may focus on insignificant details and ignore important aspects of a situation.

The frontal lobes are also involved in language, particularly linking words to form sentences, and in motor functions, such as moving the arms, legs, and mouth.

The **temporal lobes**, located below and to the side of each frontal lobe on the right and left sides of the brain, contain essential areas for memory and play a major role in language and emotions. They help people understand words, speak, read, write, and connect words with their meanings. They allow people to recognize objects and to relate appropriate emotions to objects and events. When the temporal lobes aren't working properly, people may have difficulty recognizing and responding appropriately to emotions.

Which lobe — and which part of the lobe — is affected first determines which symptoms appear initially. For example, if the disease starts in the part of the frontal lobe responsible for decision-making, then the first symptom might be trouble managing finances. If it begins in the part of the temporal lobe that connects emotions to objects, then the first symptom might be an inability to recognize potentially dangerous objects — a person might reach for a snake or plunge a hand into boiling water, for example.

In rare cases, frontotemporal disorders can begin in a different part of the brain. For example, when a person has early movement problems, such as falls, swallowing difficulties, or muscle stiffness, the disease may be beginning in the movement parts of the brain, such as the basal ganglia or brainstem.

What Do the Terms Mean?

One of the challenges shared by people living with these disorders, families, clinicians, and researchers is confusion about how to classify and label frontotemporal disorders. A diagnosis by one doctor may be called something else by a second, and the same condition or disorder may be referred to by a different name by a pathologist who examines the brain after death.

Here, we use the term frontotemporal dementia and the abbreviation FTD to refer to frontotemporal dementia. We also cover other frontotemporal disorders, including corticobasal syndrome, progressive supranuclear palsy, and primary progressive aphasia.

Frontotemporal disorders are not a single brain disease but rather a family of neurodegenerative diseases. FTD is perhaps the most commonly used term, but other widely used terms include frontotemporal lobar degeneration, frontal lobe disorders, and frontotemporal degeneration. FTD also used to be referred to as Pick's disease, and some doctors still use the term today. It's important to note that with some frontotemporal disorders, the primary symptoms are problems with speech or movement, rather than dementia symptoms. Physicians and psychologists diagnose the different forms of FTD based on a person's symptoms as well as the results of brain scans and genetic tests.

Types and Symptoms of Frontotemporal Disorders

Frontotemporal disorders can be grouped into three types, defined by the earliest symptoms. In the early stages, it can be hard to know which type of FTD a person has because symptoms and the order in which they appear can vary from one person to another. Also, the same symptoms can appear across different disorders and vary from one stage of the disease to the next as different parts of the brain are affected. In general, changes in the frontal lobe are associated with behavioral symptoms, while changes in the temporal lobe affect language and emotions. Because it is rare and not well understood, FTD is sometimes misdiagnosed as Alzheimer's disease or other types of dementia.

Symptoms of FTD are often misunderstood. Family members and friends may think that a person is misbehaving, leading to anger and conflict. It is important to understand that people with these disorders cannot control their behaviors and other symptoms and (before they are diagnosed) lack any awareness of their illness.

Behavioral Variant Frontotemporal Dementia

The most common FTD, behavioral variant frontotemporal dementia (bvFTD), involves changes in personality, behavior, and judgment. People with bvFTD can act strangely around other people, resulting in embarrassing social situations. Often, they don't know or care that their behavior is unusual and don't show any consideration for the feelings of others. People with this disorder may or may not have problems with cognition or memory.

Symptoms can include:

- Problems planning and sequencing
- Difficulty prioritizing tasks or activities
- Repeating the same activity or saying the same word over and over
- Acting impulsively or saying or doing inappropriate things without considering how others perceive the behavior
- Becoming uninterested in family or activities they used to care about
- Displaying flat, exaggerated, or improper emotions that seem disconnected from the situation
- Difficulty reading social signals, seeming to lack empathy
- Compulsive eating or taking food from others' plates

Over time, language and/or movement problems may occur, and the person living with bvFTD will need more care and supervision.

Primary Progressive Aphasia

Primary progressive aphasia (PPA) involves changes in the ability to communicate — to use language to speak, read, write, and understand what others are saying. This includes difficulty using or understanding words (aphasia) and difficulty speaking properly (e.g., slurred speech). People with PPA may have one or both of these symptoms. They may eventually become unable to speak.

Many people with PPA develop symptoms of dementia. Problems with memory, reasoning, and judgment are not apparent at first but can develop over time. Some people with PPA may experience significant behavioral changes, similar to those seen in bvFTD, as the disease progresses.

There are three types of PPA, categorized by the language problems that appear first.

- **Semantic PPA:** A person slowly loses the ability to understand single words and sometimes to recognize the faces of familiar people and common objects.
- **Agrammatic PPA:** A person has more and more trouble speaking and may omit words that link nouns and verbs, such as *to*, *from*, and *the*. Eventually, the person may no longer be able to speak at all. The person may later develop movement symptoms similar to those seen in corticobasal syndrome (see below).
- **Logopenic PPA:** A person has trouble finding the right words during a conversation but can understand words and sentences. The person does not have problems with grammar. Logopenic PPA is usually accompanied by the hallmark brain changes seen in Alzheimer's.



Researchers do not fully understand the biological basis of the different types of PPA. But they hope one day to link specific language problems with the changes in the brain that cause them.

Movement Disorders

Two rare neurological movement disorders associated with FTD, corticobasal syndrome and progressive supranuclear palsy, occur when the parts of the brain that control movement are damaged. The disorders may affect thinking and language abilities, too.

- **Corticobasal syndrome (CBS)** can be caused by corticobasal degeneration — a gradual atrophy (shrinking) and loss of nerve cells in specific parts of the brain. This causes progressive loss of the ability to control movement, typically beginning around age 60. The most prominent symptom may be apraxia, the inability to use the hands or arms to perform a movement despite normal strength, such as difficulty fastening buttons or operating small appliances. Other symptoms can include muscle rigidity and difficulty swallowing. Symptoms may appear first on one side of the body, but eventually both sides are affected. Occasionally, a person with corticobasal syndrome first has language problems or trouble orienting objects in space and later develops movement symptoms. Not everyone who has corticobasal syndrome has problems with memory, cognition, language, or behavior.
- **Progressive supranuclear palsy (PSP)** typically causes problems with balance and walking. People with the disorder typically move slowly, experience unexplained falls, lose facial expression, and have body stiffness, especially in the neck and upper body — symptoms similar to those of Parkinson's disease. A hallmark sign of this disorder is trouble with eye movements, particularly looking down. These symptoms may give the face a fixed stare. Problems with behavior, language, problem solving, and judgment can also develop.

Other movement-related types of FTD include frontotemporal dementia with parkinsonism and frontotemporal dementia with amyotrophic lateral sclerosis (FTD-ALS).



- **Frontotemporal dementia with parkinsonism** is usually caused by CBS or PSP, and in rare cases can be an inherited disease caused by a genetic variant in the tau protein also called the *MAPT* gene. Symptoms are similar to those of Parkinson's and include slowed movement, stiffness, balance problems, and changes in behavior or language.
- **FTD-ALS**, also called FTD with motor neuron disease, is a combination of bvFTD and ALS, with the latter sometimes known as Lou Gehrig's disease. In addition to the behavioral and/or language changes seen in bvFTD, people with FTD-ALS experience the progressive muscle weakness, fine jerks, and wiggling in muscles (fasciculation) seen with ALS. Symptoms of either disease may appear first, with other symptoms developing over time. Changes in certain genes have been found in some people with FTD-ALS.

What Causes FTD?

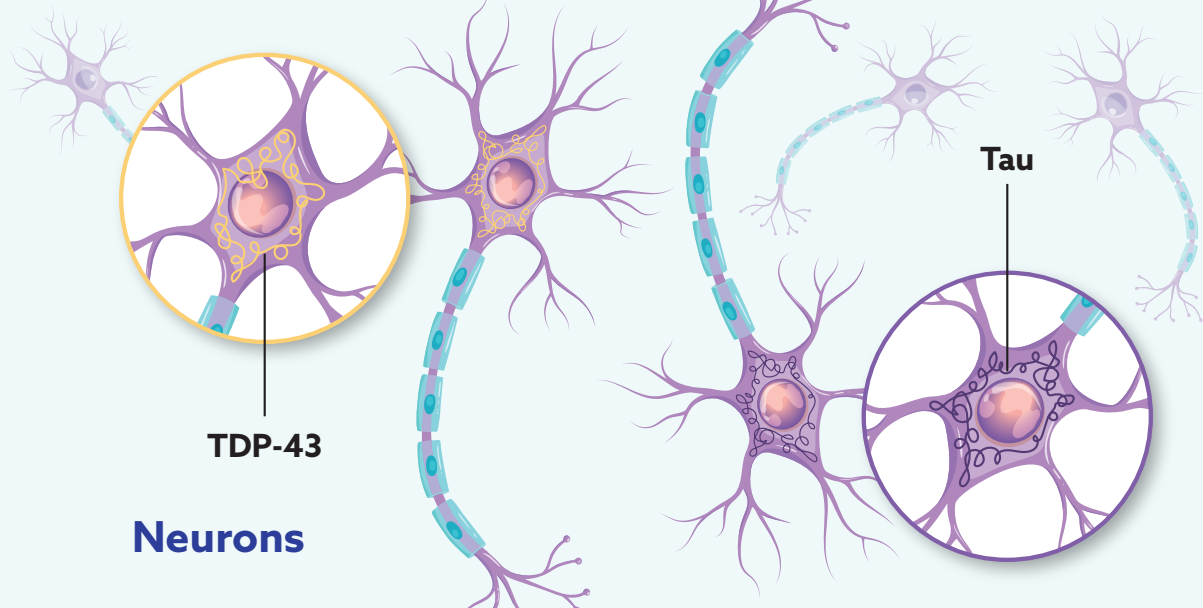
Scientists are beginning to understand the biological and genetic basis for the changes observed in brain cells that lead to FTD.

Scientists describe FTD using the patterns of change in the brain seen in an autopsy after death. These changes include loss of neurons and abnormal amounts or forms of proteins called tau and TDP-43. These proteins occur naturally in the body and help cells function properly. When the proteins don't work right, for reasons not yet fully understood, neurons in specific brain regions are damaged.

People can have changes in their brain associated with multiple forms of dementia. For example, people with corticobasal syndrome and logopenic PPA may also have beta-amyloid plaques in their brain, which are associated with Alzheimer's.

In most cases, the cause of FTD is unknown. Individuals with a family history of FTD are more likely to develop FTD.

FTD that runs in a family is often related to variants (permanent changes) in certain genes. Genes are basic units of heredity that tell cells how to make the proteins the body needs to function. Even small changes in a gene may produce an abnormal protein, which can lead to changes in the brain and, eventually, disease. An estimated one-third of FTD cases are inherited, meaning the genetic variant is passed from parent to child. These genetic variants can often be identified through genetic testing.



Neurons

Scientists have discovered several different genes that, when changed or mutated, can lead to FTD:

- ***MAPT* gene** — A change in this gene causes abnormalities in tau proteins, which then form tangles inside neurons and ultimately leads to the destruction of brain cells. Inheriting a variant in this gene means a person will almost surely develop a frontotemporal disorder, usually bvFTD, but the exact age of onset and symptoms cannot be predicted.
- ***GRN* gene** — A change in this gene can lead to lower production of the protein progranulin, which in turn causes another protein, TDP-43, to go awry in brain cells. This can lead to frontotemporal disorders, with bvFTD being the most common. The *GRN* gene can cause different symptoms in different family members and cause the disease to begin at different ages.
- ***C9orf72* gene** — An unusual change in this gene appears to be the most common genetic abnormality in familial frontotemporal disorders and familial ALS. This variant can cause a frontotemporal disorder, ALS, or both conditions.

In recent years, researchers have discovered several other genetic changes that lead to rare familial types of frontotemporal disorders. These other variants account for less than 5% of all cases of FTD.

Scientists are continuing to study these genes and to search for other genes and proteins, as well as nongenetic risk factors, that may play a role in FTD. They are trying to understand, for example, how mutations in a single gene lead to different types of FTD in members of the same family. Environmental factors that may influence risk for developing the disorders are also being examined.

Families affected by inherited and familial forms of FTD can help scientists advance research by participating in clinical studies and trials. For more information, talk with a health care professional or visit the [Alzheimers.gov](http://www.alzheimers.gov/clinical-trials) clinical trials page at www.alzheimers.gov/clinical-trials.



How Is FTD Diagnosed?

FTD can be hard to diagnose because the symptoms resemble those of other conditions. For example, bvFTD is sometimes misdiagnosed as a mood disorder, such as depression. To make matters more confusing, a person can have both FTD and another type of dementia, such as Alzheimer's. Also, because these disorders are rare, clinicians may be unfamiliar with the signs and symptoms.

To help diagnose FTD, a doctor may:

- Perform an exam and ask about symptoms
- Review personal and family medical history
- Use laboratory tests to help rule out other conditions, including Alzheimer's
- Order genetic testing
- Conduct evaluations to assess behavior, memory, thinking, language skills, and physical function
- Order imaging of the brain

A psychiatric evaluation can help determine if depression or another mental health condition is causing or contributing to the condition. A diagnosis of FTD can only be confirmed by genetic tests (in familial cases) or a brain autopsy after death.

Brain Imaging and FTD

Different types of brain imaging may be used. A magnetic resonance imaging (MRI) scan shows changes in the size and shape of the brain, including the frontal and temporal lobes. It may reveal other potentially treatable causes of the person's symptoms, such as a stroke or tumor. In the early stage of disease, the MRI may appear normal. Other types of imaging, such as positron emission tomography (PET) may be useful. PET scans measure activity in the brain by monitoring blood flow, glucose usage, and oxygen usage. Other PET scans can help rule out a diagnosis of Alzheimer's.

Getting the wrong diagnosis can be frustrating. Without an accurate diagnosis, people with FTD and related disorders may not get appropriate treatment to manage their symptoms. Families may not get the help they need. See page 28 for medical centers that can help people with FTD get diagnosis and treatment.

Researchers are studying ways to diagnose FTD earlier and more accurately and to distinguish them from other types of dementia. One area of research involves biomarkers, such as proteins or other substances in the blood or cerebrospinal fluid, which can be used to help diagnose FTD and measure disease progression. Results from biomarker tests also can help rule out other causes of dementia. For example, a blood test for Alzheimer's could help establish whether the cause of dementia is likely to be Alzheimer's or something else. Researchers are also exploring ways to improve brain imaging and neuropsychological testing.

How Is FTD Treated and Managed?

So far, there is no cure for FTD and no way to slow down or prevent it. However, there are ways to manage symptoms. A team of specialists — doctors; nurses; and speech, physical, and occupational therapists — familiar with these disorders can help guide treatment.

Managing Behavior Changes in FTD

Behavior changes associated with bvFTD can upset and frustrate family members and other caregivers. Understanding changes in personality and behavior and knowing how to respond can reduce frustration and help provide the best care for a person with FTD. Finding expert assistance can be helpful in learning the best strategies to manage behavior changes. See page 33 for information on resources that can connect patients and families with experts to provide support.

Managing behavioral symptoms can involve several approaches. Here are some strategies to consider:

- Try to accept rather than challenge someone with behavioral symptoms. Arguing or reasoning will not help, because they cannot control their behaviors or see that their behaviors are unusual or upsetting to others. Instead, be as sensitive as possible and understand that it's the illness "talking."
- Take a "timeout" when frustrated — take deep breaths, count to 10, or leave the room for a few minutes.

- To deal with apathy, limit choices and offer specific options. Open-ended questions, such as “What do you want to do today?” are more difficult to answer than specific ones, such as “Do you want to go to the park or for a walk?”
- Maintain a regular schedule, reduce distractions, and modify the environment to reduce confusion and improve the person’s sleep.
- If compulsive eating is an issue, consider supervising eating, limiting food choices, locking cabinets and the refrigerator, and distracting the person with other activities.

To ensure the safety of the person and their family, caregivers may have to take on new responsibilities or arrange care that was not needed before.

Medications are available to treat certain behavioral symptoms. Antidepressants called selective serotonin reuptake inhibitors are commonly prescribed to treat social disinhibition and impulsive behavior. People with aggression or delusions sometimes take low doses of antipsychotic medications. If a particular medication is not working, a doctor may try another. Always consult a doctor before changing, adding, or stopping a drug or supplement.

Treating Language Problems in Primary Progressive Aphasia and FTD

Treatment of PPA has two goals: maintaining language skills and using new tools and other ways to communicate. Treatment tailored to a person’s specific language problem and stage of PPA generally works best. Since language ability declines over time in people with PPA, different strategies may be needed as the illness progresses. The following may help:

- Use a communication notebook (an album of photos labeled with names of people and objects), gestures, and drawings to communicate without talking.
- Store lists of words or phrases in a computer or smartphone to point to.
- Speak slowly and clearly, use simple sentences, wait for responses, and ask for clarification if needed.
- Work with a speech-language pathologist familiar with PPA to determine the best tools and strategies to use. Note that many speech-language pathologists are trained to treat aphasia caused by stroke, which requires different strategies from those used with PPA.

MARY ANN'S STORY

Mary Ann, a television news anchor for 20 years, began having trouble reading the nightly news. At first, her doctor thought she had a vision problem, but tests showed that her eyesight was normal. Although normally creative and energetic, Mary Ann, 52, had trouble finishing assignments and voicing her ideas at staff meetings. In time, she was let go from her job. Mary Ann applied for Social Security disability benefits, which required a medical exam. Her symptoms puzzled several doctors until a neurologist diagnosed logopenic PPA. A speech therapist taught Mary Ann to use a personal digital assistant to express words and phrases. For emergencies, Mary Ann carries a card in her wallet that explains her condition.

Managing Movement Problems in FTD

Medications and physical and occupational therapy may provide modest relief for the movement symptoms of FTD and related frontotemporal disorders. A doctor who specializes in FTD can guide treatment.

For people with PSP, Parkinson's medicines may offer some temporary improvement. Physical and occupational therapy may help the person move more easily. Speech therapy can help them manage language symptoms.

For people with progressive supranuclear palsy, sometimes Parkinson's disease drugs provide temporary relief for slowness, stiffness, and balance problems. Exercises can keep the joints limber, and weighted walking aids — such as a rolling walker with brakes — can help maintain balance. Speech, vision, and swallowing difficulties usually do not respond to any drug treatment; however, antidepressants have shown modest success, and patients with uncontrollable laughing or crying often respond to medications for these problems. For people with abnormal eye movements, special glasses called prisms are sometimes prescribed.

People with FTD-ALS typically decline quickly over two to three years. During this time, physical therapy can help treat muscle symptoms, and a walker or wheelchair may be useful. Speech therapy may help a person speak more clearly at first. Later on, other ways of communicating, such as a speech synthesizer, can be used. The ALS symptoms of the disorder ultimately make it impossible to stand, walk, eat, and breathe on one's own.

Physicians, nurses, social workers, and physical, occupational, and speech therapists who are familiar with these conditions can ensure that people with movement disorders get appropriate medical treatment and that their caregivers can help them live as well as possible.

What is the Future of FTD Treatment?

Researchers are continuing to explore the biological changes in the body, including genetic variants and proteins, that lead to FTD and identify and test possible new drugs and other treatments. They also want to develop better ways, such as specialized brain imaging, to track progression, so that treatments, when they become available, can be directed to the people for whom they will work best.

Clinical trials and studies are underway to advance these efforts. People with FTD and healthy people may be able to participate. The FTD Disorders Registry is a contact and research registry for people diagnosed with frontotemporal disorders. It is open to family members, caregivers, or friends of people diagnosed with FTD. Registries help speed up trials by gathering a list of “willing and able” potential participants. People who sign up may be referred to specific studies but are not obligated to participate. See page 36 for more information about the FTD Disorders Registry.

To find out more about participating in clinical research related to FTD, talk to your health care provider or visit the [Alzheimers.gov](https://www.alzheimers.gov/clinical-trials) clinical trials page at www.alzheimers.gov/clinical-trials.

Providing Care For a Person With FTD

Caring for someone with FTD can be hard, both physically and emotionally. Caregivers may face challenges with managing medical and day-to-day care, as well as changing family and social relationships, loss of work, poor health, stress, decisions about long-term care, and end-of-life concerns.

Strategies to Help Manage FTD Symptoms

There is no cure for FTD and no way to slow it down or prevent it. However, there are ways to help manage symptoms.

- **Managing behavior changes in FTD.** Try to recognize it's the illness "talking" and accept rather than challenge people with behavioral symptoms. Arguing or reasoning with the person will not help; they cannot control their behavior or even see that their behavior is unusual or upsetting to others.
- **Treating language problems in FTD.** To help with language issues, speak slowly and clearly, use simple sentences, wait for responses, and ask for clarification if you don't understand something. Gesturing, drawing, and using an album with labeled photos of people and objects may help communicate without talking. A speech-language pathologist can help determine the best tools and strategies for a particular person.
- **Managing movement problems in FTD.** Several types of FTD cause problems with movement, including difficulty balancing, walking, and swallowing. Medications and physical

and occupational therapy may provide modest relief for the movement symptoms of FTD. A doctor who specializes in these disorders can guide treatment.

Be a Part of Finding Better Treatments

Do you know someone living with FTD? Researchers are looking for people to participate in a study to track disease progression and advance treatments.

Visit www.allftd.org to learn more.

How FTD and Related Disorders Affect Families

People with FTD and their families must cope with changing relationships, especially as symptoms get worse. Spouses or partners may find themselves not only taking on caregiving responsibilities, but also household responsibilities that their partner can no longer perform. Children may suffer the gradual

JUSTIN'S STORY

After Justin graduated from college, he went home to live with his parents. It didn't take long for him to notice personality changes in his 50-year-old mother, a successful executive. She became more childlike and had trouble finishing household chores. By the time she was diagnosed with bvFTD, Justin's relationship with his mother had deteriorated. Learning about the disorder helped Justin understand and accept the changes he was seeing in his mother.

loss of a parent before they reach adulthood. Family members and friends may feel alienated or embarrassed by the person's behavior. Life at home can become very stressful.

How FTD Can Affect Work

People living with FTD may have difficulty with basic work skills, such as organizing, planning, and following through on tasks. Activities that were easy before might take much longer or become impossible. People may lose their jobs because they can no longer perform them. As a result, the caregiver might need to take a second job to make ends meet, or reduce their hours, or even quit working to provide care and run the household.

An employment attorney can offer information and advice about employee benefits, family leave, and disability if needed. Workers diagnosed with FTD can qualify for Social Security disability benefits through the "Compassionate Allowances" program (800-772-1213), a program that helps individuals with certain serious conditions access benefits quickly.

FTD Caregiver Health and Support

Caring for someone with FTD presents unique challenges. Many caregivers face declines in their own health while caring for a person with FTD or a related disorder. To stay healthy, caregivers can:

- Get regular health care.
- Ask family and friends for help with child care, errands, and other tasks.

- Spend time doing enjoyable activities, away from the demands of caregiving. Arrange for respite care — short-term caregiving services that give the regular caregiver a break — or take the person to an adult day care center, a safe, supervised environment for adults with dementia or other disabilities.
- Join a support group for caregivers of people with frontotemporal disorders. Such groups can be a valuable resource to share experiences and tips with others who may be in a similar situation. Nonprofit organizations and community organizations may provide online or in-person support groups.

The organizations listed in the Resources section can help with information about caregiver services and support.

Long-Term Care For People With FTD

For many caregivers, there comes a point when they can no longer take care of the person with FTD without professional help. The caregiving demands are simply too great, and the person may need around-the-clock care. As the disease progresses, caregivers may want to get home health care services or look for a residential care facility, such as a group home, assisted living facility, or nursing home.



The decision to move the person with FTD to a care facility can be difficult, but it can also give caregivers peace of mind to know that the person is safe and getting good care. The decreased level of stress may also improve the caregivers' relationship with their loved one.

End-of-Life Concerns

People with FTD typically live six to eight years with their conditions, sometimes longer, sometimes less. Most people die of problems related to advanced disease. For example, as movement skills decline, a person can have trouble swallowing, leading to aspiration pneumonia, in which food or fluid gets into the lungs and causes infection. People with balance problems may fall and seriously injure themselves.



It is difficult, but important, to plan for the end of life. Legal documents, such as a will, living will, and durable powers of attorney for health care and finances, should be created or updated as soon as possible after a diagnosis of FTD or a related disorder. Early on, many people can understand and participate in legal decisions. But as their illness progresses, it becomes harder to make such decisions.

A doctor who knows about frontotemporal disorders can help determine the person's mental capacity. An attorney who specializes in elder law, disabilities, or estate planning can provide legal advice, prepare documents, and make financial arrangements for the caregiving spouse or partner and dependent children. If necessary, the person's access to finances can be reduced or eliminated.

Conclusion

It is impossible to predict the exact course of FTD and other frontotemporal disorders. These disorders are not easy to live with, but with help, people can meet the challenges and prepare for the future. Getting an early, accurate diagnosis and the right medical team are crucial first steps. Researchers and clinicians are working toward a deeper understanding of frontotemporal disorders and better diagnosis and treatment to help people manage these difficult conditions. You can help pave the way for new and more effective treatments for people living with FTD by joining a clinical trial.



Where to Find FTD Diagnosis and Treatment

Because FTD and other frontotemporal disorders are rare and can cause similar symptoms to other conditions, it can be difficult to get an accurate diagnosis. The centers below have clinicians and researchers focused specifically on FTD.

Case Western Reserve University

Department of Neurology
Cleveland, OH
216-464-6445

Cleveland Clinic Nevada

Lou Ruvo Center for Brain Health
Las Vegas, NV
702-483-6000

Columbia-Presbyterian Medical Center

Department of Neurology
New York, NY
646-426-3876

Emory University

School of Medicine
Atlanta, GA
404-727-6123

Houston Methodist Hospital

Frontotemporal Degeneration Unit

Houston, TX

713-441-7650

Indiana University School of Medicine

Indiana Alzheimer's Disease Center

Indianapolis, IN

317-963-5500

Johns Hopkins University School of Medicine

Frontotemporal Dementia and Young-Onset Dementias Clinic

Baltimore, MD

410-955-5147

Massachusetts General Hospital

Frontotemporal Disorders Unit

Boston, MA

617-726-1728

Mayo Clinic

Department of Neurology

Rochester, MN

507-538-3270

Jacksonville, FL

904-953-0853

Phoenix or Scottsdale, AZ

800-446-2279

Mount Sinai

Icahn School of Medicine
New York, NY
212-241-6500

Nantz National Alzheimer Center

Houston, TX
Phone: 713-441-1150

Northwestern University Feinberg School of Medicine

Mesulam Center for Cognitive Neurology and Alzheimer's Disease
Chicago, IL
312-908-9339

University of Alabama, Birmingham

Neurology Department, Division of Memory Disorders
Birmingham, AL
205-996-3679

University of British Columbia

Faculty of Medicine
Vancouver, BC, Canada
604-822-2421

University of California, Los Angeles

Neurobehavior Clinic
Los Angeles, CA
310-794-1195

University of California, San Diego

Shiley-Marcos Alzheimer's Disease Research Center
La Jolla, CA
858-822-4800

University of California, San Francisco

Memory and Aging Center
San Francisco, CA
415-353-2057

University of Colorado, Denver

Anschutz Medical Campus
Aurora, CO
303-724-5000

University of Michigan

Department of Neurology
Ann Arbor, MI
734-936-9020

University of North Carolina, Chapel Hill

Division of Memory and Cognitive Disorders
Chapel Hill, NC
984-974-4401

University of Pennsylvania Health System

Penn Frontotemporal Degeneration Center
Philadelphia, PA
215-349-5863

University of Toronto

Tanz Centre for Research in Neurodegenerative Diseases
Toronto, ON, Canada
416-507-6879

University of Washington

School of Medicine

Seattle, WA

206-543-6116

Vanderbilt University

Department of Neurology

Nashville, TN

615-936-0060

Washington University School of Medicine in St. Louis

Department of Neurology

St. Louis, MO

314-362-1408

Where Can I Find More Information About FTD?

Information may be available from the following organizations:

Federal Government

Alzheimers.gov

800-438-4380

adear@nia.nih.gov

www.alzheimers.gov

National Institute on Aging (NIA)

**Alzheimer's and related Dementias Education
and Referral (ADEAR) Center**

800-438-4380

adear@nia.nih.gov

www.nia.nih.gov/alzheimers

MedlinePlus

www.medlineplus.gov

National Institute of Mental Health (NIMH)

866-615-6464

nimhinfo@nih.gov

www.nimh.nih.gov

National Institute of Neurological Disorders and Stroke (NINDS)

800-352-9424

www.ninds.nih.gov

Eldercare Locator

800-677-1116

eldercarelocator@USAgings.org

eldercare.acl.gov

Social Security Administration

800-772-1213

800-325-0778 (TTY)

www.ssa.gov

FTD-Related Associations and Organizations

ALS Association

800-782-4747

alsinfo@als.org

www.als.org

Alzheimer's Association

800-272-3900

www.alz.org

Alzheimer's Foundation of America

866-232-8484

info@alzfdn.org

www.alzfdn.org

Association for Frontotemporal Degeneration (AFTD)

866-507-7222

info@theaftd.org

www.theaftd.org

BrightFocus Foundation

800-437-2423

info@brightfocus.org

www.brightfocus.org/alzheimers

CurePSP

800-457-4777

info@curepsp.org

www.psp.org

John Douglas French Alzheimer's Foundation

310-445-4650

cheryl@jdfaf.org

www.jdfaf.org

National Aphasia Association

800-922-4622

naa@aphasia.org

www.aphasia.org

National Organization for Rare Disorders (NORD)

203-744-0100

rarediseases.org

Caregiver Support

Association for Frontotemporal Degeneration Support Groups

866-507-7222

info@theaftd.org

www.theaftd.org/aftd-support-groups

Family Caregiver Alliance

800-445-8106

www.caregiver.org

FTD Disorders Registry

888-840-9980

manager@FTDregistry.org

ftdregistry.org

FTD Support Forum

www.ftdsupportforum.com

National Academy of Elder Law Attorneys, Inc.

703-942-5711

naela@naela.org

www.naela.org

Well Spouse Association

732-577-8899

info@wellspouse.org

www.wellspouse.org

The National Institute on Aging (NIA) and the National Institute of Neurological Disorders and Stroke (NINDS) are part of the National Institutes of Health, the nation's medical research agency — supporting scientific studies that turn discovery into health.

NIA leads the federal government effort conducting and supporting research on aging and the health and well-being of older people. NIA's Alzheimer's and related Dementias Education and Referral (ADEAR) Center offers information and publications on dementia and caregiving for families, caregivers, and professionals.

NINDS is the nation's leading funder of research on the brain and nervous system. The mission of NINDS is to seek fundamental knowledge about the brain and nervous system and to use that knowledge to reduce the burden of neurological disease for all people.



National Institutes of Health

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